0.1 Average velocity and acceleration

$$v_{ave} = \frac{\Delta x}{\Delta t}$$
$$a_{ave} = \frac{\Delta v}{\Delta t}$$

0.2 Instantaneous velocity and acceleration

$$v = \frac{dx}{dt}$$
$$a = \frac{dv}{dt}$$

0.3 Constant acceleration

$$v = v_0 + at$$
$$x = v_0 t + \frac{1}{2}at^2$$
$$v^2 - v_0^2 = 2ax$$

0.4 Relative motion

$$\vec{v_1} = \vec{v_2} + \vec{v_3}$$

0.5 Projectile motion

$$x = v_0 \cos(\theta)t, \qquad y = v_0 \sin(\theta)t - \frac{1}{2}gt^2$$
$$v_x = v_0 \cos(\theta), \qquad v_y = v_0 \sin(\theta) - gt$$
$$y = x \tan(\theta) - \frac{g}{2v_0^2 \cos^2(\theta)}x^2$$

0.6 Quadratic formula

If $at^2 + bt + c = 0$, then:

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$